공학수학(2)

14.3 Double Integrals in Polar Coordinates

Review of Polar Coordinates

We know that from Figure 1 that the polar coordinates (r, θ) of a point are related to the rectangular coordinates (x, y) by the equations

$$r^2 = x^2 + y^2 \qquad x = r\cos\theta \qquad y = r\sin\theta$$

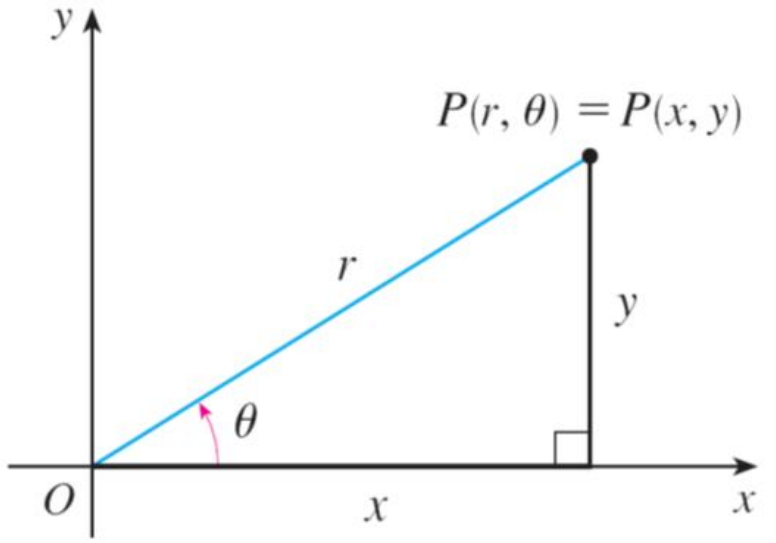
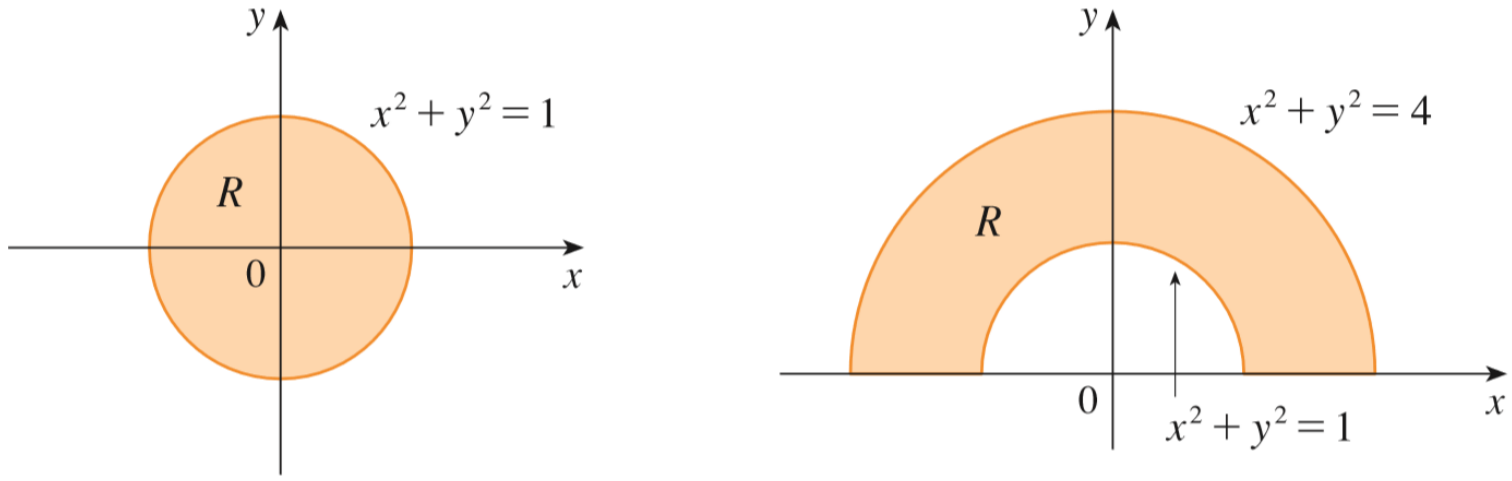


Figure 1

Review of Polar Coordinates

Equations of circles centered at the origin are particularly simple in polar coordinates. The unit circle has equation $r = 1$; the region enclosed by this circle is shown in Figure 2(a). Figure 2(b) illustrates another region that is conveniently described in polar coordinates.



(a) $R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$

(b) $R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$

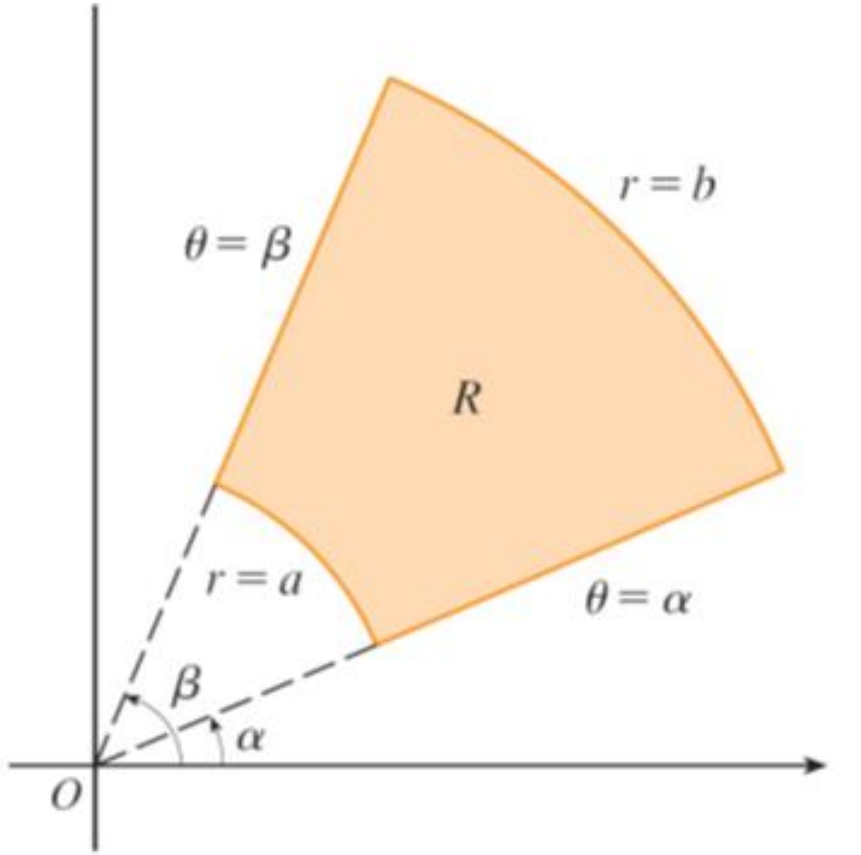
Figure 2

Double Integrals in Polar Coordinates

The regions in Figure 2 are special cases of a **polar rectangle**

$$R = \{(r, \theta) \mid a \le r \le b, \ \alpha \le \theta \le \beta\}$$

which is shown in Figure 3.



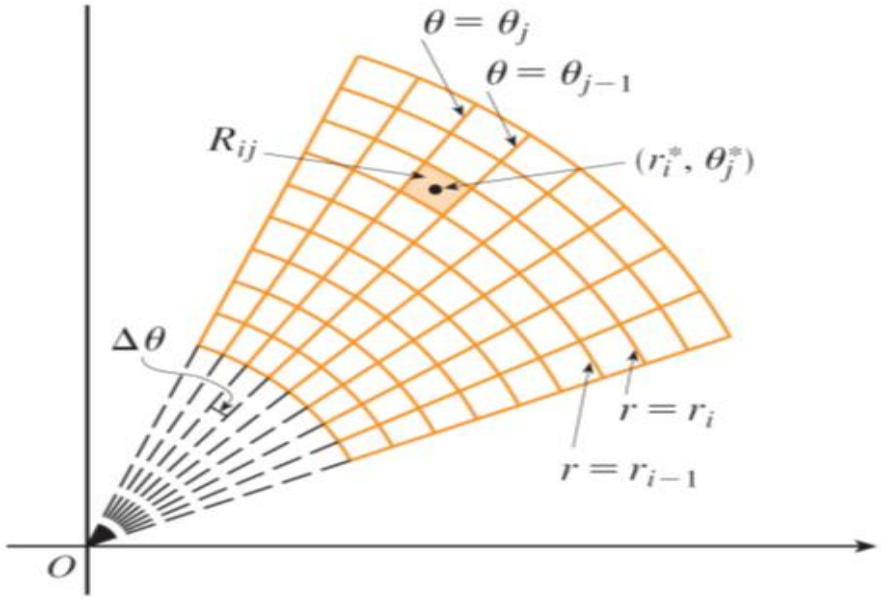
Polar rectangle

Figure 3

Double Integrals in Polar Coordinates

In order to compute the double integral $\iint_R f(x, y)\, dA$, where R is a polar rectangle, we divide the interval $[a, b]$ into m subintervals $[r_{i-1}, r_i]$ of equal width $\Delta r = \dfrac{(b-a)}{m}$ and we divide the interval $[\alpha, \beta]$ into n subintervals $[\theta_{j-1}, \theta_j]$ of equal width $\Delta\theta = \dfrac{(\beta-\alpha)}{n}$.

Then the circles $r = r_i$ and the rays $\theta = \theta_j$ divic the polar rectangle R into the small polar rectangles R_{ij} shown in Figure 4.



Dividing R into polar subrectangles

Figure 4

Double Integrals in Polar Coordinates

The “center” of the polar subrectangle

$$R_{ij} = \left\{ (r,\ \theta) \middle| r_{i-1} \le r \le r_i,\ \theta_{j-1} \le \theta \le \theta_j \right\}$$

has polar coordinates

$$r_i^* = \tfrac{1}{2}(r_{i-1} + r_i) \qquad \theta_j^* = \tfrac{1}{2}(\theta_{j-1} + \theta_j)$$

We compute the area of R_{ij} using the fact that the area of a sector of a circle with radius r and central angle θ is $\frac{1}{2} r^2 \theta$.

Double Integrals in Polar Coordinates

Subtracting the areas of two such sectors, each of which has central angle $\Delta\theta = \theta_j - \theta_{j-1}$, we find that the area of R_{ij} is

$$\begin{aligned}\Delta A_i &= \tfrac{1}{2} r_i^2 \Delta\theta - \tfrac{1}{2} r_{i-1}^2 \Delta\theta = \tfrac{1}{2}(r_i^2 - r_{i-1}^2)\Delta\theta \\ &= \tfrac{1}{2}(r_i + r_{i-1})(r_i - r_{i-1})\,\Delta\theta = r_i^*\ \Delta r \Delta\theta\end{aligned}$$

Although we have defined the double integral $\iint_R f(x,y)\,dA$ in terms of ordinary rectangles, it can be shown that, for continuous functions f, we always obtain the same answer using polar rectangles.

Double Integrals in Polar Coordinates

The rectangular coordinates of the center of R_{ij} are $(r_i^* \cos\theta_j^*, r_i^* \sin\theta_j^*)$, so a typical Riemann sum is

$$\sum_{i=1}^{m}\sum_{j=1}^{n} f(r_i^* \cos\theta_j^*,\ r_i^* \sin\theta_j^*)\ \Delta A_i = \sum_{i=1}^{m}\sum_{j=1}^{n} f(r_i^* \cos\theta_j^*,\ r_i^* \sin\theta_j^*)\ r_i^*\ \Delta r\ \Delta\theta \qquad 1$$

If we write $g(r, \theta) = rf(r\cos\theta, r\sin\theta)$, then the Riemann sum in Equation 1 can be written as

$$\sum_{i=1}^{m}\sum_{j=1}^{n} g(r_i^*,\ \theta_j^*)\ \Delta r\ \Delta\theta$$

which is a Riemann sum for the double integral

$$\int_{\alpha}^{\beta}\int_{a}^{b} g(r,\ \theta)\ dr\ d\theta$$

Double Integrals in Polar Coordinates

Therefore we have

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(r_i^*\cos\theta_j^*,\ r_i^*\sin\theta_j^*)\,\Delta A_i$$

$$= \lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} g(r_i^*,\ \theta_j^*)\,\Delta r\ \Delta\theta = \int_\alpha^\beta\int_a^b g(r,\ \theta)\,dr\ d\theta$$

$$= \int_\alpha^\beta\int_a^b f(r\cos\theta,\ r\sin\theta)\,r\ dr\,d\theta$$

2 Change to Polar Coordinates in a Double Integral If f is continuous on a polar rectangle R given by $0 \le a \le r \le b,\ \alpha \le \theta \le \beta$, where $0 \le \beta - \alpha \le 2\pi$, then

$$\iint_R f(x,y)\,dA = \int_\alpha^\beta\int_a^b f(r\cos\theta,\ r\sin\theta)\,r\ dr\,d\theta$$

Double Integrals in Polar Coordinates

The formula in (2) says that we convert from rectangular to polar coordinates in a double integral by writing $x = r\cos\theta$ and $y = r\sin\theta$, using the appropriate limits of integration for r and θ, and replacing dA by $r\,dr\,d\theta$.

Be careful not to forget the additional factor r on the right side of Formula 2.

A classical method for remembering this is shown in Figure 5, where the "infinitesimal" polar rectangle can be thought of as an ordinary rectangle with dimensions $r\,d\theta$ and dr and therefore has "area" $dA = r\,dr\,d\theta$.

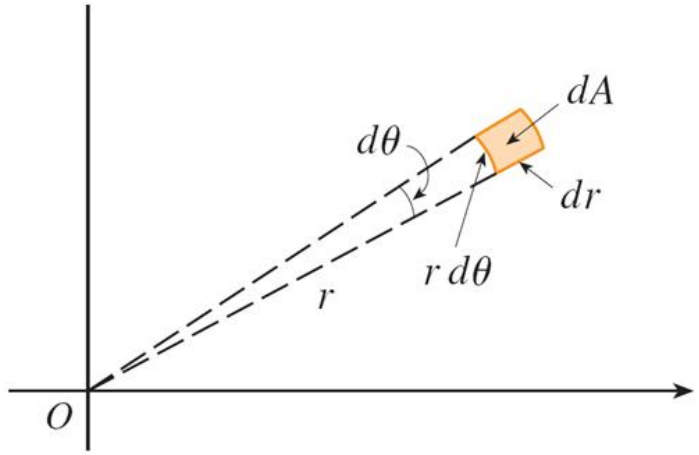


Figure 5

Example 1

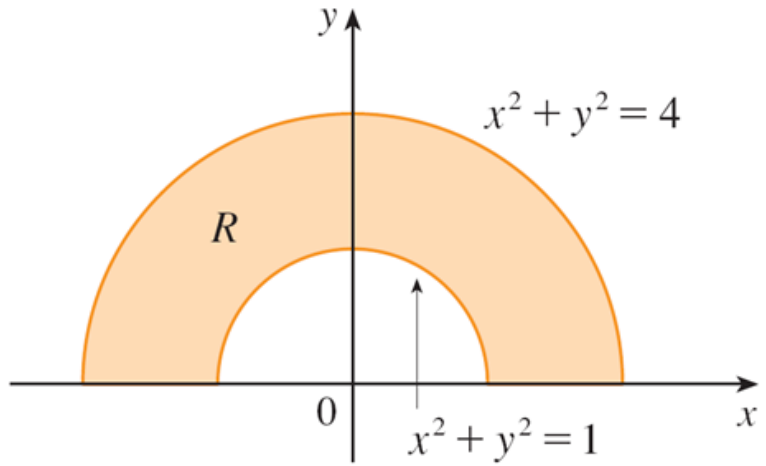
Evaluate $\iint_R (3x + 4y^2)\, dA$, where R is the region in the upper half-plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Solution:

The region R can be described as

$$R = \{(x, y) \mid y \geq 0,\ 1 \leq x^2 + y^2 \leq 4\}$$

It is the half-ring shown in Figure 2(b), and in polar coordinates it is given by $1 \leq r \leq 2$, $0 \leq \theta \leq \pi$.



$$R = \{(r, \theta) \mid 1 \leq r \leq 2,\ 0 \leq \theta \leq \pi\}$$

Figure 2(b)

Example 2

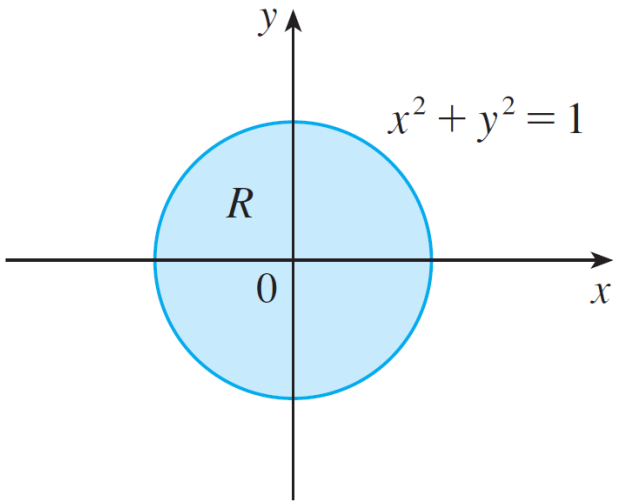
Find the volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$

Solution:

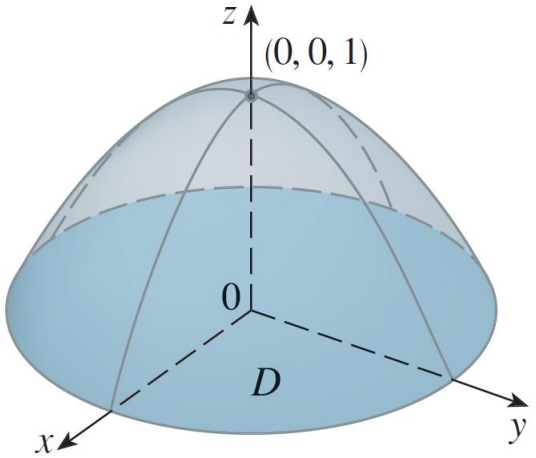
If we put $z = 0$ in the equation of the paraboloid, we get $x^2 + y^2 = 1$.

Example 2

This means that the plane intersects the paraboloid in the circle $x^2 + y^2 = 1$, so the solid lies under the paraboloid and above the circular disk D given by $x^2 + y^2 \leq 1$ [see Figures 6 and 1(a)].



$$R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$$



Double Integrals in Polar Coordinates

What we have done so far can be extended to the more complicated type of region shown in Figure 8. In fact, by combining Formula 2 with

$$\iint_D f(x,\ y)\ dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,\ y)\ dx\ dy$$

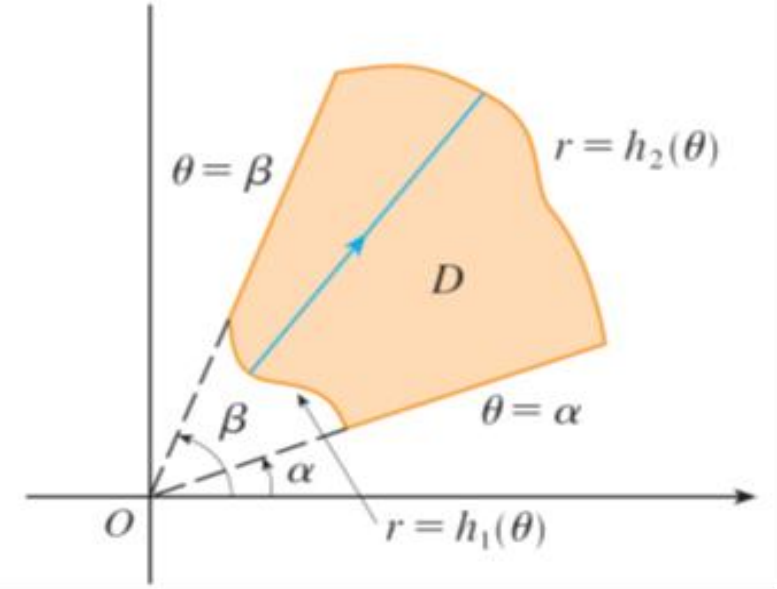
where D is a type II region, we obtain the following formula.

3 If f is continuous on a polar region of the form

$$D = \left\{(r,\ \theta) \middle| \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\right\}$$

then

$$\iint_D f(x,\ y)\ dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta,\ r\sin\theta)\ r\ dr\ d\theta$$



$$D = \left\{(r,\theta) \middle| \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\right\}$$

Figure 8

Double Integrals in Polar Coordinates

In particular, taking $f(x, y) = 1$, $h_1(\theta) = 0$, and $h_2(\theta) = h(\theta)$ in this formula, we see that the area of the region D bounded by $\theta = \alpha$, $\theta = \beta$, and $r = h(\theta)$ is

$$A(D) = \iint_D 1 \, dA$$

$$= \int_\alpha^\beta \int_0^{h(\theta)} r \; dr \; d\theta$$

$$= \int_\alpha^\beta \left[\frac{r^2}{2} \right]_0^{h(\theta)} d\theta$$

$$= \int_\alpha^\beta \tfrac{1}{2} \left[h(\theta) \right]^2 d\theta$$

Example 5

Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy-plane, and inside the cylinder $x^2 + y^2 = 2x$.

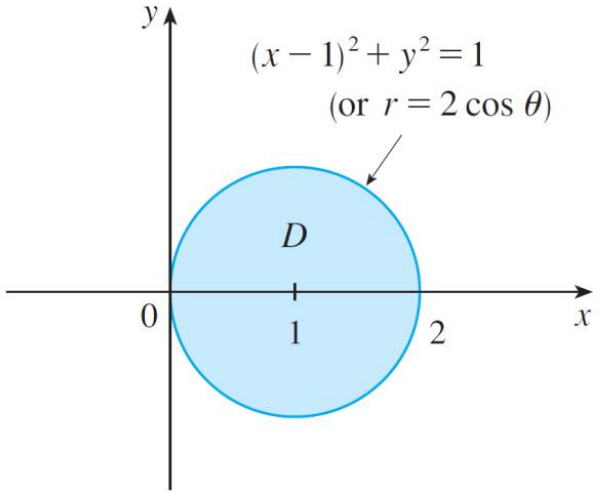
Solution:

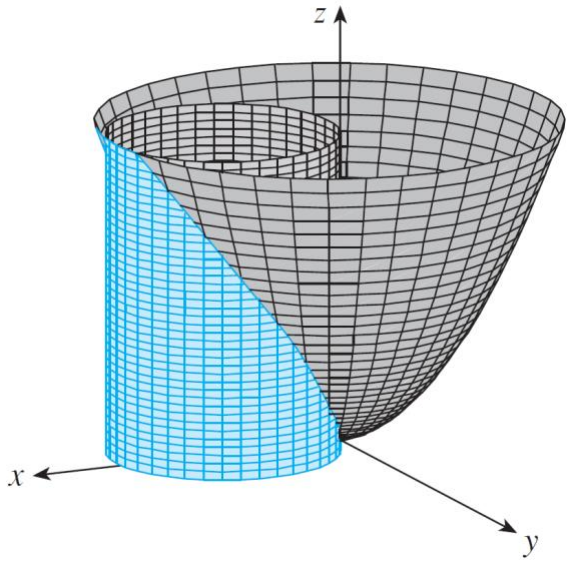
The solid lies above the disk D whose boundary circle has equation $x^2 + y^2 = 2x$ or, after completing the square,

$$(x - 1)^2 + y^2 = 1$$

Example 5

See Figures 10 and 11. In polar coordinates we have $x^2 + y^2 = r^2$ and $x = r\cos\theta$, so the boundary circle becomes $r^2 = 2r\cos\theta$, or $r = 2\cos\theta$.





Thus the disk D is given by

$$D = \{(r, \theta) \mid -\pi/2 \leq \theta \leq \pi/2,\ 0 \leq r \leq 2\cos\theta\}$$